

Queue ADT, Operations & Array Implementation

Q. Define Queue as an Abstract Data Type (ADT). Explain its standard operations and demonstrate its implementation using an Array with a relevant code snippet.

> **Definition to Remember**

> A **Queue** is a linear data structure in which the elements are inserted at one end and removed from the other end. It follows the First In First Out (FIFO) principle.

1. Queue as an ADT

As an Abstract Data Type, a Queue manages a collection of elements and restricts access to two specific ends:

- **Enqueue:** Inserting an element at the rear end.

2. Standard Operations

Operation	Description	Condition
Enqueue	Inserting an element at the rear end.	if (rear == -1 & front == -1) { rear = 0; front = 0; }
Dequeue	Removing an element from the front end.	if (front == -1 & rear == -1) { return -1; }

3. Array Implementation in C

Two pointers (`front` and `rear`) are initialized to `-1`.

```
```c
#include <stdio.h>
#define SIZE 100
int queue[SIZE];
int front = -1, rear = -1;

void enqueue(int value) {
 if (rear == -1 & front == -1) {
 front = 0;
 rear = 0;
 } else if (rear < front) {
 rear = 0;
 }
 queue[rear] = value;
 rear++;
}

int dequeue() {
 if (front == -1 & rear == -1) {
 return -1;
 }
 int val = queue[front];
 front++;
 return val;
}
```
```

4. Advantages and Limitations

- **Advantage:** Array implementation is simple and fast ($O(1)$ time complexity for enqueue/dequeue).

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

Complete Executable C Code: Linear Queue Array Implementation

```
```c
```

```
#define MAX 5
```

```
struct Queue {
```

```
void initQueue(struct Queue *q) {
```

```
int isFull(struct Queue *q) {
```

```
int isEmpty(struct Queue *q) {
```

```
return
```

```
void enqueue(struct Queue *q, int val) {
```

```
if (isFu
```

```
int dequeue(struct Queue *q) {
```

```
if (isEr
```

```
void display(struct Queue *q) {
```

```
if (isEr
```

```
int main() {
```

```
struct
```

```
enqueue(&q, 10);
```

```
enque
```

```
printf("Dequeued: %d\n", dequeue(&q));
```

```
display
```

```
return 0;
```

```
}
```